Cuckoo: An IE Free Rider Hatched by Massive Nutrition in LLM's Nest

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Abstract

Massive high-quality data, both pre-training raw texts and post-training annotations, have been carefully prepared to incubate advanced large language models (LLMs). In contrast, for information extraction (IE), pre-training data, such as BIO-tagged sequences, are hard to scale up. We show that IE models can act as free riders on LLM resources by reframing next-token *prediction* into *extraction* for tokens already present in the context. Specifically, our proposed next tokens extraction (NTE) paradigm learns a versatile IE model, *Cuckoo*[1], with 102.6M extractive data converted from LLM's pre-training and post-training data. Under the few-shot setting, Cuckoo adapts effectively to traditional and complex instruction-following IE with better performance than existing pre-trained IE models. As a free rider, Cuckoo can naturally evolve with the ongoing advancements in LLM data preparation, benefiting from improvements in LLM training pipelines without additional manual effort.[2]

1 Introduction

The biggest lesson researchers have learned from training large language models (LLMs) (Wang et al., 2023b; Touvron et al., 2023; Achiam et al., 2023; Groeneveld et al., 2024; Dubey et al., 2024; Team et al., 2024) is the power of massive and high-quality data (Kaplan et al., 2020; Hernandez et al., 2021). Although pre-training information extraction (IE) models (Huang et al., 2021; Tedeschi and Navigli, 2022; Lu et al., 2022; Li et al., 2023; Bogdanov et al., 2024; Peng et al., 2024) has once been a popular topic before the rise of general LLMs, the relative scarcity of automated annotations has limited the further development of this domain. Consequently, more and more researchers have accepted LLMs as backbone models for IE tasks (Agrawal et al., 2022; Wang et al., 2023a; Xu et al., 2024b).

The primary reason for the temporary lag in IE pre-training is the stricter format requirements for data collection compared to those for LLMs. The paradigm for learning LLMs, the next token prediction (NTP), can utilize every token in the sentence as an annotation. In contrast, IE pre-training always requires spans annotated with label names. While certain platforms provide massive annotations, such as Page Links in Wikipedia (Balasuriya et al., 2009; Ding et al., 2021; Han et al., 2018; Tedeschi and Navigli, 2022), they are still much less efficient than NTP. To illustrate the gap, Multinerd (Tedeschi and Navigli, 2022) takes multiple processing efforts to collect 164K English named entity recognition (NER) instances from Wikipedia and Wikinews, while NTP can easily gather trillions of tokens from raw texts as supervision.

This paper proposes a frustratingly simple yet effective way to scale up IE pre-training. We suggest that IE pre-training can simply be a free rider on the LLM's training resources by learning on exactly the same pre-training and post-training datasets. We modify NTP to next tokens extraction (NTE), using BIO tags for next tokens that can be extracted from the input context as shown in Figure 1. With the instruction-following ability learned in post-training, one can adjust the prompt to instruct NTE-based taggers to perform different IE tasks.

Specialized for IE, NTE has three advantages over NTP. 1) Parameter Efficiency, NTP requires extra parameters to store knowledge to generate tokens not in the input context, while NTE concentrates only on tagging input tokens. Thus, NTE-based IE taggers can have better parameter efficiency than NTP-based LLMs, fitting it to smaller models like RoBERTa (Liu et al., 2019). 2) Inference Efficiency, NTE taggers are not only smaller because of the parameter efficiency but can also extract multiple tokens with the BIO scheme in

[1]Cuckoo is known for laying its eggs in other birds' nests, tricking them into raising its chicks.

[2]Open Cuckoo: https://github.com/KomeijiForce/Cuckoo

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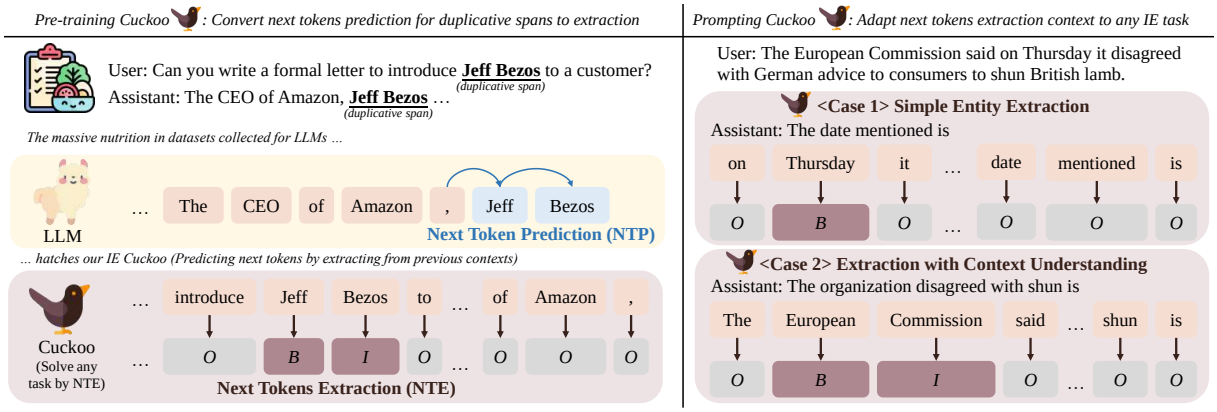
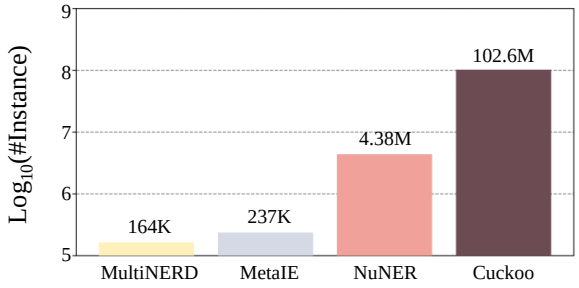


Figure 1: Cuckoo takes a free ride on LLM resources (e.g., C4 and TuluV3 (Lambert et al., 2024)) by formalizing next token prediction for duplicative spans as extraction in the BIO paradigm. During the inference, the prompts can be adjusted to different extractive tasks, making Cuckoo a versatile IE model.



	MultiNERD	MetaIE	NuNER	Cuckoo
Source	Annotation	Synthesis	Synthesis	Raw Text
Cost	$0	$$$$	$$$$	$0
Diversity	NER Label	IE Label	NER Label	Any Context
Perspective	Grow by Wikipedia	Grow by More Synthesis	Grow by More Synthesis	Evolve with LLM Training

Figure 2: Comparison of scale, cost, and diversity among different IE pre-training datasets. Our data collection for Cuckoo is free by converting LLM's learning resources, which forces the tagger to learn from diverse contexts. Cuckoo can also evolve with the data collection for LLM's post-training.

one forward pass. 3) Transferability, NTE taggers can easily adapt to IE tasks, which are typically annotated in the same BIO scheme.

With NTE, we easily collect 100M pre-training instances from C4[3] (Raffel et al., 2020), a popular pre-training dataset, and 2.6M chat-formatted instances from TuluV3 post-training dataset (Lambert et al., 2024) to endow the model with instruction-following ability. We continually train a RoBERTa tagger on massive NTE data, which results in our *Cuckoo* model, a free rider with a training paradigm similar to NTP on training resources for LLMs. We present the comparison of scale, cost and diversity with other IE pre-training datasets in Figure 2.

[3]We estimate the English part of C4 can be transformed into 5B instances, we only take 100M (2%) for experiment efficiency.

We follow the few-shot adaptation evaluation in previous works (Tedeschi and Navigli, 2022; Bogdanov et al., 2024) to benchmark Cuckoo, which shows that Cuckoo is as versatile as LLMs in extractive tasks. Training with few-shot data, Cuckoo can quickly understand different kinds of NER labels, free text questions in machine reading comprehension, and complex instructions, to perform precise extraction. With overwhelming advantages in data scale, Cuckoo outperforms models pre-trained on massive human-annotated or LLM-synthesized datasets by a large margin.

Finally, we analyze to show 1) Cuckoo can evolve with the data collection for LLM's post-training data; 2) in-context tagging ability emerges in Cuckoo just like in-context learning in LLMs; and 3) Cuckoo scales up by the increasing number of our constructed NTE data.

2 Background

Information Extraction Information extraction (IE) is one of the most fundamental applications in natural language processing. IE systems take the user's requirement (e.g., defined by a label text, a question, or an instruction) and extract spans of several tokens from input texts. The two most frequent categories of IE targets are entity and relation, which structure many IE tasks, such as named entity recognition (Sang and Meulder, 2003), relation extraction (Carreras and Màrquez, 2004), event extraction (Walker et al., 2006), and others (Carreras and Màrquez, 2005; Pontiki et al., 2014; Xu et al., 2020). A crucial challenge to modern IE systems is the growing number of IE targets (e.g., various label names) in the open world, which are scarce in annotation and require IE systems for quick trans-

fer learning. Thus, many works have collected massive automated IE annotations to pre-train IE models (Ding et al., 2021; Tedeschi and Navigli, 2022; Li et al., 2023; Bogdanov et al., 2024; Peng et al., 2024), which shows benefits in transferring to low-resource IE targets.

Large Language Model The biggest game-changer for natural language processing in all domains is the large language model (LLM) (Wang et al., 2023b; Touvron et al., 2023; Achiam et al., 2023; Groeneveld et al., 2024; Dubey et al., 2024; Team et al., 2024). Learning on trillions of tokens for pre-training and post-training, LLMs have shown surprisingly strong performance on all kinds of tasks (Achiam et al., 2023). Next token prediction, the paradigm behind the success of LLMs, supports exploiting every token in raw texts as the annotation to strengthen the model's capability. Consequently, many IE researchers have turned toward LLMs (Agrawal et al., 2022; Wang et al., 2023a; Xu et al., 2024b) to use them as strategic information extractors with planning (Huang et al., 2024; Kim et al., 2024) and chain-of-thoughts (Wei et al., 2022; Ma et al., 2023).

Pre-training Paradigm: IE v.s. LLM The rise of LLMs has challenged the meaningfulness of IE pre-training with an overwhelmingly larger number of annotations. The lagging of IE pre-training can be attributed to the relatively high format requirement for IE annotation like labels in Wikipedia links. This paper shows IE pre-training can take a free ride on LLM's NTP paradigm to unleash the power of massive pre-training.

3 Our Cuckoo

3.1 Next Tokens Extraction

The learning paradigm for LLMs is next token prediction (NTP), which calculates the representation of a context $[x_1, x_2, \cdots, x_t]$ to output a probability distribution p_{t+1} of the next token x_{t+1} over all potential tokens in the LLM's vocabulary. The prediction p_{t+1} is optimized by the cross entropy loss to maximize its value on x_{t+1}.

We modify NTP into next tokens extraction (NTE) for cases that the span of next n tokens $[x_{t+1}, \cdots, x_{t+n}]$ already exist in the context $[x_1, x_2, \cdots, x_t]$, such that $[x_{k+1}, \cdots, x_{k+n}] = [x_{t+1}, \cdots, x_{t+n}] (1 \leq k \leq t-n)$. When we detect such (t, k, n), we annotate IE tags for the context as $[l_1, l_2, \cdots, l_t]$ following a BIO scheme. We first set all tags l to *O*. As there can be multiple k for t, for each k, we set l_k to *B* and $[l_{k+1}, \cdots, l_{k+n}]$ to *I*. The high-level idea of NTE is to replace prediction by extraction for duplicative spans that appear multiple times in the context.

NTE thus allows IE pre-training to directly exploit NTP datasets for LLM training, which significantly broadens the potential training data. During the inference, one can adjust the prompts of an NTE-based tagger to instruct it to perform different kinds of extractive tasks. Recall the strengths mentioned for NTE in the introduction, NTE specialized for IE has advantages in parameter efficiency, inference efficiency, and adaptability over NTP.

3.2 Massive Nutrition for Cuckoo

Pre-training and Post-Training With NTP-to-NTE conversion, we can simply copy the two training stages for LLMs, to perform pre-training and post-training for NTE-based IE taggers. Pre-training learns raw texts while post-training learns instruction-following dialogues between the user and the IE assistant. During pre-training, we annotate BIO tag sequences based on all (t, k, n) triplets, assuming the multiple appearances of the same span of tokens indicate a certain level of extractive relation (Gu et al., 2021). For post-training, we suppose the extraction should focus on the texts provided by users so we only keep (t, k, n) triplets that k falls in the user's request and t falls in the assistant's response.

Then, we select the resources for pre-training and post-training. While the NTE framework allows us to exhaust all kinds of resources, we use only one dataset for each stage for experiment efficiency. For pre-training, we select the popular C4 (CommonCrawl) dataset (Raffel et al., 2020), which contains 4B passages and is commonly used to pre-train LLMs. For post-training, we use the most advanced TuluV3 (Lambert et al., 2024) dataset with 939K instruction-following interactions between the user and the assistant.

To further boost the experiment efficiency, we first collect noun phrases parsed by SpaCy[4], filtering stop words or punctuations. Then we collect 5% of the rest spans (no overlapping) that are duplicative to produce NTE instances. On C4, we keep the first 100M NTE instances transformed from the raw texts. On TuluV3, we transform all post-training interactions into the NTE format, re-

[4] https://spacy.io/

sulting in 2.6M instances. We also sample 5% spans not existing in their previous contexts, whose NTE labels are annotated by all *O* as negative cases.

With the 102.6M instances, we continually pre-train a `roberta-large` model (Liu et al., 2019) as the BIO tagger for NTE, optimized by AdamW (Loshchilov and Hutter, 2019) with learning rate initialized to 10^{-5}. The batch size is set to 64, taking about 1.6M steps for the optimization.

3.3 Statistics

Besides the huge scale, we analyze other key statistics of our massive NTE dataset to investigate its efficiency in learning various IE targets. Our investigation is respectively done on the two pre-training and post-training data splits.

How "extractive" are the data? An obvious concern on the NTE dataset is whether the automated annotations reflect real extractive relations. We prompt the advanced LLM, `gpt-4o` (Achiam et al., 2023), to identify whether NTE data establish real extractive relations. The responses on 20K sampled data show 93.39% pre-training data and 96.20% post-training data contain extractive relations, which shows the high data efficiency of the annotation strategy.

How diverse are the data? The data is extremely diverse by containing any duplicative spans in a broad domain. We find around 28M unique spans in C4 and 0.4M in TuluV3, which is combined with highly diverse contexts in C4 and TuluV3. Our dataset covers various span lengths (maximally 40 words) and context lengths (maximally 512 words). The proportion of span with ≥ 4 tokens is 4.52%, which seems small but still contains 4.6M spans because of the large scale of our dataset. Our context length is also more diverse than previous IE pre-training resources (Tedeschi and Navigli, 2022; Bogdanov et al., 2024; Peng et al., 2024) where data only have one or two sentences as context.

What is the conversion rate? The conversion rate from a sentence to an NTE instance is 332% for C4 and 235% for TuluV3. This is highly efficient in comparison with traditional IE pre-training datasets relying on scarce links or expensive synthesis. The full C4 dataset can be transformed into 5B NTE instances. However, the efficiency is still relatively lower than NTP. Only 4.06% tokens in pre-training and 4.14% tokens in post-training are used for NTE tagger learning, which indicates the supervision from LLM resources can be further augmented.

Level	Example
Basic	Organization
Query	Which organization launched the campaign?
Instruction	Organization (Disambiguation: The organization entity must be a subject of any active action in the context.)

Table 1: IE targets of different understanding levels.

4 Experiments

Different from previous evaluation procedures that enumerate IE tasks (Lu et al., 2022; Paolini et al., 2021; Peng et al., 2024), our evaluation splits IE tasks into different levels of understanding the IE target. Specifically, the three levels are 1) Basic IE, understanding a single label text for an entity or a relation, such as named entity recognition. 2) Query-based IE, understanding a sentence-level query, such as machine reading comprehension (MRC). 3) Instruction-following IE, understanding complex extractive instructions like LLMs.

Examples of different understanding level are enumerated in Table 1. We expect that Cuckoo will be comparable to traditional IE pre-training on Basic IE as most popular label texts have been enumerated by LLM synthesis (Bogdanov et al., 2024; Peng et al., 2024). Cuckoo's advantage over traditional IE pre-training is on query-based and instruction-following IE, which requires understanding more complex IE targets.

4.1 Benchmark and Evaluation

Following the high-level evaluation objective, we use several traditional benchmarks for each level of IE ability. Method and benchmark details are included in Appendices A and B.

Basic IE benchmarks the understanding of simple labels for entity and relation. We include 4 named entity recognition datasets (CoNLL03 (Sang and Meulder, 2003), BioNLP2004 (Collier and Kim, 2004), MIT-Restaurant/Moive (Ushio and Camacho-Collados, 2021)) and 2 relation extraction datasets (CoNLL04 (Carreras and Màrquez, 2004) and ADE (Gurulingappa et al., 2012)).

Query-based IE requires the understanding of more complex sentence-level semantics of the IE target. We thus include 3 machine reading comprehension datasets (SQuAD (Rajpurkar et al., 2016),

SQuAD-V2 (Rajpurkar et al., 2018), DROP (Dua et al., 2019)). We filter out non-extractive questions in DROP.

Instruction-following IE is a feature of LLMs when they are applied for IE. Users can include detailed requirements for the IE target in the prompt, which is hard for traditional IE systems that only understand simple label texts. However, instruction-following IE currently lacks of benchmarks[5]. Based on the real role of instruction in IE, we apply rules and a strong LLM, `GPT-4o`, to synthesize 3 instruction-following IE by modifying traditional benchmarks. 1) **Disambiguation**, we write a definition instruction for 3 ambiguous types, (*"Organization"* in CoNLL2003, *"Protein"* in BioNLP2004, *"Location"* in MIT-Restaurant), such as *"Disambiguation: The organization entity must be a subject of any active action in the context."*. We use `GPT-4o` to filter out entities that no longer meet the IE target, resulting in a new instruction-following IE benchmark. 2) **Preference**, there are different ground truth answers in machine reading comprehension like "Bruno Mars", "Mars". However, one might prefer the longer or the shorter answer. Thus, we modify the SQuAD dataset with 3 instructions with a preference for "Longer answer", "Shorter answer", "Concise answer (Answer with no extra words)"[6]. This filtering modification is automated by functions with no LLM involved. 3) **Miscellaneous**, we write 3 instructions to define the *"Miscellaneous"* entity type in CoNLL2003, MIT-Restaurant, and MIT-Movie. In practice, we clarify the existing miscellaneous type for CoNLL2003 and combine 3 minority types as miscellaneous for MIT-Restaurant and MIT-Movie. We calculate metrics only on miscellaneous entities to evaluate whether the model can understand the scope definitions.

The evaluation continues with the model's few-shot adaptability. The model will be fine-tuned on a few examples in the training set and then evaluated on the test set. For basic IE, we will have 5 shots for each entity/relation category. For query-based IE, we will have 32 training examples. For instruction-following IE, the definition of few-shot follows the original dataset. We include more details for the construction of instruction-following IE benchmark in Appendix B.

We benchmark IE performance with the traditional F1 score. For Basic IE, it refers to the Micro F1 for labeled entity spans. In Query-based IE, the F1 score refers to the maximal word-level F1 between the answer and one of the ground truths. Instruction-following IE benchmarks follow the metric of the original datasets.

4.2 Baselines and Variants

We incorporate baselines into our experiments to validate our two main claims. 1) NTE is a paradigm that can scale up the data resources for IE pre-training, which learns taggers with better few-shot adaptability, especially in instruction-following. 2) NTE is a more efficient paradigm than NTP for IE, which results in significantly stronger extractive ability of NTE-based taggers than NTP-based LMs.

For 1), we include previous IE pre-training resources to compare their pre-training effects with our NTE-based dataset. These resources include,

- **MultiNERD** (Tedeschi and Navigli, 2022) is a NER pre-training dataset based on Wikipedia and Wikinews, which contains 164K instances in the English split with 17 label names. The annotations are from community contributors.
- **NuNER** (Bogdanov et al., 2024) is a massive NER pre-training dataset synthesized by ChatGPT-3.5 (OpenAI, 2023) on massive raw texts. NuNER has 4.38M instances with 273K unique label names.
- **MetaIE** (Peng et al., 2024) is a massive IE pre-training dataset synthesized by ChatGPT-3.5 and 4 with a broader coverage than simple NER. The LLMs are prompted to enumerate possible important information for entities and relations. MetaIE includes 237K IE instances with 31K unique label names.

In addition to resources using annotations for label names, we also consider machine reading comprehension as a pre-training task for IE, as it can be viewed as query-based IE. We thus include,

- **MRQA** (Fisch et al., 2019) is a collection of machine reading comprehension data that extracts an answer from a passage for a question in each instance. We exclude SQuAD as it is used for benchmarking, which remains 488K instances.

For 2), we use the same resources for Cuckoo (C4+TuluV3) to continually pre-train an OPT model (Zhang et al., 2022) in the same parameter scale ($\sim$ 300M) as the base model RoBERTa

[5]Existing InstructIE benchmarks (Jiao et al., 2023; Gui et al., 2024) concentrate more on using instruction for traditional IE than instruction-awareness.

[6]This means when "Los Angeles", "the US" and "US" all exist in the answer candidates, "the US" will be removed but "Los Angeles" will be kept.

	Method	Named Entity Recognition					Relation Extraction		
		CoNLL2003	BioNLP2004	MIT-Restaurant	MIT-Movie	Avg.	CoNLL2004	ADE	Avg.
zero	Cuckoo	35.38	23.62	8.11	9.06	19.04	48.95	34.67	41.81
	Rainbow Cuckoo	38.56	22.07	35.38	29.53	31.39	53.81	62.01	57.91
few-shot	OPT-C4-TuluV3	50.24	39.76	58.91	56.33	50.56	47.14	45.66	46.40
	RoBERTa	33.75	32.91	62.15	58.32	46.80	34.16	2.15	18.15
	MRQA	72.45	55.93	68.68	66.26	65.83	66.23	67.44	66.84
	Cuckoo	73.60	57.00	67.63	67.12	**66.34**	69.57	71.70	**70.63**
	Only Pre-train	72.46	55.87	66.87	67.23	65.61	68.14	69.39	68.77
	Only Post-train	72.80	56.10	66.02	67.10	65.51	68.66	69.75	69.21
	MultiNERD†	66.78	54.62	64.16	66.30	60.59	57.52	45.10	51.31
	NuNER†	74.15	56.36	68.57	64.88	65.99	65.12	63.71	64.42
	MetaIE†	71.33	55.63	70.08	65.23	65.57	64.81	64.40	64.61
	Rainbow Cuckoo †	79.94	58.39	70.30	67.00	**68.91**	70.47	76.05	**73.26**

Table 2: Performance comparison on Basic IE Tasks. †: In-domain Transfer. (Transfer learning on the same task and format as the pre-training stage.)

of Cuckoo. We select OPT because its NTP pre-training resource has covered the one for RoBERTa (Liu et al., 2019; Zhang et al., 2022), which eliminates the attribution of Cuckoo's advantage to a better base model (RoBERTa).

For the ablation study, we include the variants of Cuckoo, which only use the LLM's pre-training (C4) or post-training (TuluV3) resource for IE pre-training. These two variants aim to demonstrate the contributions of both stages to justify the imitation of the LLM's training pipeline.

Rainbow Cuckoo Finally, we incorporate a strong variant combining more post-training resources, *Rainbow Cuckoo*. Rainbow Cuckoo extends the post-training resource from only TuluV3 to merging multiple datasets including samples from MultiNERD, NuNER, MetaIE, and MRQA, which aims to exploit all possible resources to further boost the IE pre-training.

Zero-shot Performance is also evaluated on our Cuckoo and its variant Rainbow Cuckoo to demonstrate the direct performance after the IE pre-training on LLM's resources.

4.3 Basic IE

The performance on basic IE tasks is presented in Table 2. Our two main claims are supported by the experiment results,

1) Cuckoo outperforms all baselines using different IE pre-training resources on both entity and relation extraction. Among the baselines, the best-performing ones are NuNER for entity and MRQA for relation, which they specialize in. Cuckoo overwhelms the baselines with a much larger pre-training data scale. As Cuckoo with only the raw texts from C4 (pre-training) has already achieved comparable or better performance than baselines, the conversion to NTE shows strong data efficiency on raw texts.

	Method	SQuAD	SQuAD-V2	DROP	Avg.
zero	Cuckoo	48.82	49.16	38.41	45.46
	Rainbow Cuckoo	82.79	57.67	61.62	67.36
few-shot	OPT-C4-TuluV3	39.80	53.81	31.00	41.54
	RoBERTa	31.86	48.55	9.16	29.86
	MultiNERD	42.85	50.99	30.12	41.32
	NuNER	61.60	52.67	37.37	50.55
	MetaIE	74.59	62.54	30.73	55.95
	Cuckoo	77.47	64.06	54.25	**65.26**
	Only Pre-train	75.64	63.36	52.81	63.94
	Only Post-train	77.05	62.39	54.80	64.75
	MRQA†	80.07	66.22	54.46	66.92
	Rainbow Cuckoo †	86.57	69.41	64.64	**73.54**

Table 3: Performance comparison on Query-based IE Tasks. †: In-domain Transfer.

2) The NTE pre-trained RoBERTa (Cuckoo) outperforms the NTP pre-trained OPT, which validates our intuition that language models can be more parameter efficient by focusing on extraction.

Besides the validation of our main claims, we also have more discoveries from the performance of variants. The first observation is that both pre-training and post-training datasets contribute to adaptability. In basic IE tasks, the massive raw texts in C4 contribute more than the curated post-training data in TuluV3, which indicates the basic IE tasks are simple enough to be well transferred by learning without annotations. The Rainbow Cuckoo shows Cuckoo can be further enhanced with merging more post-training resources, demonstrating significantly strong IE ability.

	Method Base Task	Disamb. NER	Prefer. MRC	Misc. NER
zero	Cuckoo	13.88	35.56	2.93
	Rainbow Cuckoo	21.93	60.81	14.62
few-shot	OPT-C4-TuluV3	28.56	53.68	37.19
	RoBERTa	12.29	6.04	9.71
	MultiNERD	31.71†	30.84	44.68†
	NuNER	31.40†	51.01	44.32†
	MetaIE	29.77†	56.12	47.35†
	Cuckoo	**34.97**	62.53	**49.17**
	Only Pre-train	32.21	59.64	46.05
	Only Post-train	34.28	**64.37**	47.28
	MRQA	29.33	66.83†	48.67
	Rainbow Cuckoo	**37.75**†	**70.95**†	**51.86**†

Table 4: Performance comparison on Instruction-following IE tasks for disambiguation (Disamb.), preference (Prefer.), and miscellaneous (Misc.). †: In-domain Transfer.

4.4 Query-based IE

We present the performance of models on query-based IE (MRC) in Table 3. Among out-of-domain models, Cuckoo significantly outperforms other models pre-trained on basic IE tasks, rivaling the model pre-trained on the in-domain MRQA dataset. The result exhibits the benefit of NTE to pre-train in a wild and diverse raw text distribution, contrasting the fixed templates in basic IE pre-training. Post-training resources show a more significant contribution to query-based than basic IE tasks as queries in MRC require higher instruction awareness. Merging MRQA into the pre-training, Rainbow Cuckoo shows a significant advantage over using only MRQA via unifying all kinds of pre-training resources by the NTE paradigm.

4.5 Instruction-following IE

Table 4 demonstrates the instruction-following ability of different IE models. The zero-shot performance implies that the task requires a higher-level understanding of IE instructions. Cuckoo once again significantly outperforms other models except for an in-domain case (MRQA on MRC-based preference instruction testing) and widens the gap, showing its strong adaption to new instructions with the following ability learned from LLM pre-training resources. Post-training data contribute the most to the ability to follow instructions, playing the same role as for LLMs. Occasionally, learning only post-training data outperforms the full Cuckoo. Rainbow Cuckoo, with a large amount of post-training supervision, once again significantly boosts the performance.

Method	Long	Short	AnsSim ↓	DualEM
Cuckoo	57.84	51.39	**40.48**	11.67
MRQA	62.61	61.05	48.17	12.32
Rainbow Cuckoo	**67.20**	**63.67**	44.58	**18.95**

Table 5: Detailed analysis on the instruction-following ability of IE models with preference as an example.

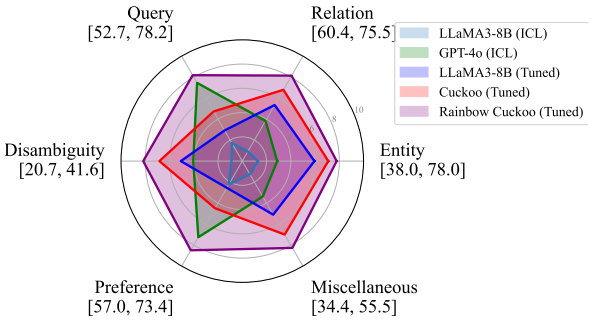


Figure 3: The performance comparison between Cuckoo and LLMs on few-shot IE performance.

Cuckoo reacts to instruction. We provide a deeper investigation of Cuckoo's reactions to instructions. Specifically, we test the preference instructions for the longest and shortest answers, which will lead to different answers. We fine-tune pre-trained IE models with few shots for both the longest and the shortest answers and then test their instruction-following ability. For evaluation, we use answer similarity (AnsSim) between outputs from two instructions, where higher similarity indicates less instruction-awareness. We also use dual exact matching (DualEM) as a strict metric to evaluate whether the model correctly reacts to both instructions. AnsSim calculates the word-level F1 score between answers from two instructions and DualEM refers to the model accuracy to produce both answers correctly. Table 5 shows that the MRQA model is no longer significantly better than Cuckoo on DualEM. AnsSim also indicates MRQA model to have less instruction-awareness, restraining its strong MRC ability to be applied with specific instructions. In comparison, the Rainbow Cuckoo shows a much higher advantage over the MRQA model according to the DualEM metric, demonstrating a better efficiency in applying the MRC ability to the instruction-following scenario.

4.6 Cuckoo v.s. LLMs

We extend the comparison to Cuckoo versus LLMs. We select `LLaMA-3-8B-TuluV3` and `GPT-4o` to represent the fine-tunable open-source LLMs and API-based close-source LLMs. For

LLaMA-3-8B-TuluV3, we fine-tune the LLM with the same templated data as our Cuckoo. For both LLMs, we evaluate their in-context learning IE ability based on the few shots.

We present the experiment result in Figure 3, which demonstrate that Cuckoo can outperform even fine-tuned 8B LLMs. This implicates the superior learning efficiency of NTE over NTP on IE tasks. The ICL performance of LLM significantly lags behind the fine-tuned one, restraining the performance of close-source LLMs. Finally, Rainbow Cuckoo validates itself again as the strongest few-shot IE learner even when LLMs are considered.

Efficiency The time efficiency of Cuckoo is significantly higher than LLMs thanks to the specialized learning paradigm for IE. Taking NER as an example, Cuckoo is around 60× faster than LLaMA-3-8B-TuluV3 (166.79 instance/s versus 2.75 instance/s). When the LLM is using ICL, the efficiency advantage becomes more than 50×, demonstrating the superior efficiency of Cuckoo.

5 Analyses

5.1 Evolution with LLMs

A feature of our Cuckoo is its evolution with LLM's training resources, especially for post-training data which are progressively curated by researchers (Groeneveld et al., 2024; Xu et al., 2024a; Lambert et al., 2024). In Figure 4, we plot the performance of Cuckoo post-trained by different versions of Tulu post-training datasets from V1 to V3 (Wang et al., 2023b; Ivison et al., 2023; Lambert et al., 2024) after pre-training on C4. All performances are normalized by a linear mapping from $[\mu-2\sigma, \mu+2\sigma]$[7] to $[0, 10]$ for demonstration. The result illustrates a evolution between Cuckoo and the LLMs. With each evolution in post-training data collection for LLMs, Cuckoo's performance can also be expanded in most dimensions. In the future, Cuckoo can be further improved together with the quality of LLM's training data with the free-riding feature of our NTE paradigm.

5.2 Emergence of In-context Tagging

In-context learning is an emerging skill in LLMs that adapts LLMs to new tasks with examples in the given context. We investigate whether in-context learning appears in Cuckoo, which uses a similar learning paradigm and resource as LLMs. We append 5 examples for CoNLL2003 and 1 example for SQuAD (due to context window limitation) to the context and test the in-context tagging performance of different models. In Figure 5, we find only Cuckoo able to improve (at least retain) its IE ability while other models (even pre-trained on similar tasks) show a significant drop. Thus, NTE on LLM's resources is verified to enable in-context tagging for Cuckoo. As suggested in Chan et al. (2022), the occasional burstiness in raw texts contributes to the emergence of in-context tagging in Cuckoo. While NuNER and MRQA are well formalized, they fail to learn models with in-context learning ability because of the lack of burstiness.

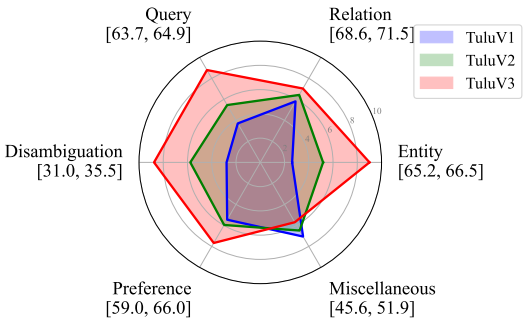


Figure 4: The evolution of Cuckoo with LLM's post-training resources. Domain $[\mu-2\sigma, \mu+2\sigma]$ is annotated under each evaluation dimension.

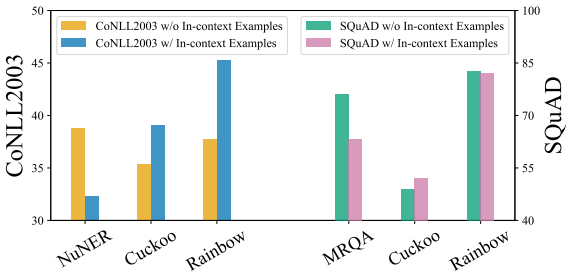


Figure 5: In-context tagging ability emerges in Cuckoo but not in IE models pre-trained by other resources.

5.3 Data Scaling Trend

Data is an important factor in the scaling law (Kaplan et al., 2020). Thus, we test the transfer learning ability of checkpoints pre-trained with different data scales to downstream tasks. We focus on the scaling law of raw texts in C4 as they are cheaper to scale up and we have discussed the evolution of Cuckoo with post-training data collection. Our investigation covers both early pre-training stages to 4.1M instances and the scaling-up to 100M.

[7] μ, σ are based on the performance of 4 Cuckoo models (before post-training, after post-training with TuluV1 to V3)

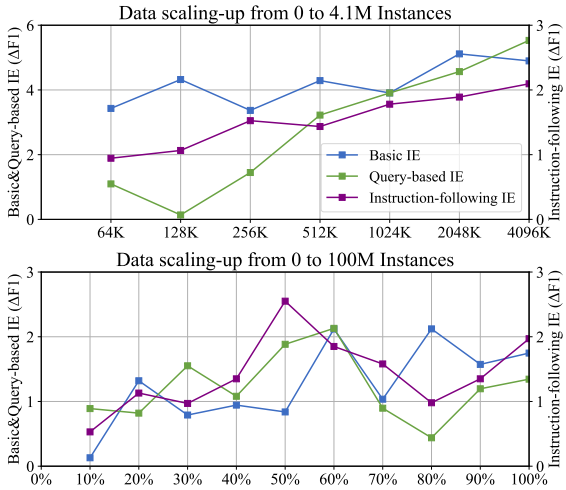


Figure 6: The data scaling trend of Cuckoo on the early 4.1M C4 instances and the massive 100M instances.

In the two subfigures of Figure 6, we plot the data scaling trend in pre-training Cuckoo. The upper figure shows a clear performance rising trend together with the increasing data amount, indicating all dimensions of IE ability are scaled-up in the early pre-training stage. In the scaling-up to 100M stage, the macroscopic trend retains its steady increase but turbulence emerges. Some intermediate checkpoints like at $50\% \sim 60\%$ data scale show a competitive performance with the fully pre-trained model. This implicates that the capacity of the small RoBERTa might meet its bound, and further improvement requires more parameters.

6 Conclusion and Future Work

This paper proposes a large-scale IE pre-training paradigm with the LLM's pre-training and post-training resources. The massive nutrition incubates a versatile Cuckoo model, which outperforms the pre-training with previous IE resources. Cuckoo can evolve with the data preparation for LLMs. Further work on Cuckoo will focus on variants in learning paradigms, datasets, and backbones.

Limitations

While Cuckoo validates the strength of NTE to take a free ride with LLM resources, our scope can be extended to several topics out of the main claims.

Label Embedding Some IE paradigms (e.g., original NuNER) learns label embeddings to efficiently label the extracted spans. As Cuckoo imitates NTP to perform NTE, its IE process requires enumerating the label names similar as the generative IE using LLMs. Matching label embedding has its efficiency advantage while generative IE allows the label texts to interact with the context, resulting in potentially better performance. Cuckoo follows the generative IE paradigm to pursue better performance based on the established success of LLMs. However, future efforted can be devoted into a label embedding version of Cuckoo, which takes the context as the label text to boost the IE efficiency.

Data Source The C4 corpus for raw text features broad coverage. However, recent progress in LLMs shows that specific sources of pre-training data (e.g., textbooks) benefit certain skills of LLMs, such as math. This paper only discusses C4 to avoid the IE performance improvement attributed to a specific data source. Future works can extend our scope to compare the effect of all kinds of resources in pre-training, which might find certain resources are superior in IE pre-training using NTE.

Backbone Variants The current scopes is designed to justify the benefit of NTE in gathering massive IE pre-training data. Thus, the comparison is biased to data quality rather than backbone models. Further exploration in backbone models include the scaling law in model size, multilingual backbone, and model architectures.

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Target	Template
Entity	**User:** [Context] Question: What is the [Label] mentioned? **Assistant:** Answer: The [Label] is
Relation (Kill)	**User:** [Context] Question: Who does [Entity] kill? **Assistant:** Answer: [Entity] kills
Relation (Live)	**User:** [Context] Question: Where does [Entity] live in? **Assistant:** Answer: [Entity] lives in
Relation (Work)	**User:** [Context] Question: Who does [Entity] work for? **Assistant:** Answer: [Entity] works for
Relation (Located)	**User:** [Context] Question: Where is [Entity] located in? **Assistant:** Answer: [Entity] is located in
Relation (Based)	**User:** [Context] Question: Where is [Entity] based in? **Assistant:** Answer: [Entity] is based in
Relation (Adverse)	**User:** [Context] Question: What is the adverse effect of [Entity]? **Assistant:** Answer: The adverse effect of [Entity] is
Query	**User:** [Context] Question: [Question] **Assistant:** Answer:
Instruction (Entity)	**User:** [Context] Question: What is the [Label] mentioned? ([Instruction]) **Assistant:** Answer: The [Label] is
Instruction (Query)	**User:** [Context] Question: [Question] ([Instruction]) **Assistant:** Answer:

Table 6: The templates used in our experiments for different tasks.

A Templates and Hyperparameters

Task Templates are included in Table 6, which are used to fine-tune NTE and NTP models like Cuckoo and LLaMA on IE tasks.

Hyperparameter All models are fully fine-tuned except for `LLaMA-3-8B-TuluV3`, which exhibits a poor performance without LoRA (Hu et al., 2022). We use a 128-dimension LoRA for `LLaMA-3-8B-TuluV3`. All fine-tuning uses AdamW (Loshchilov and Hutter, 2019) as the optimizer, learning rate initialized as 1×10^{-5} to fully fine-tune RoBERTa and OPT, and 2×10^{-4} to fine-tune the LoRA. The batch size is set to 64 for all fine-tuning.

B Benchmark Details

All results in the main experiments are an average of 3 runs on different subsets of a few shots. MRC results are evaluated on the validation split as in previous works. Instruction-following IE only focuses on the modified entity types like organization and miscellaneous.

Relation Extraction gives the ground-truth entities to extract related entities. We don't run end-to-end experiments to avoid mixing entity and relation extraction abilities.

Duplicates When an entity is extracted as multiple types in NER, we keep all of them because modern generative IE models (e.g., LLM) allow such features to fit into a broader usage. For instance, an LLM would say "Kobe Bryant" to be both a "person" and a "basketball player". For MRC, when multiple answers are extracted, we will select the answer that appears the most.

SQuAD-V2 is a special MRC dataset that contains unanswerable questions. We follow the initial evaluation to assign 1.0 F1 score to abstain for these questions and 0.0 F1 score for any answer. Adaptive training for SQuAD-V2 contains extra 32-shot unanswerable questions.

Disambiguation The 3 instructions used for disambiguation are presented in Table 7. We use the follow template to prompt `GPT-4o` for filtering.

[Instruction] Does "[Entity]" in "[Context]" satisfy the definition above? Answer "yes" or "no" only.

We manually check the filtering quality of 50 random cases for each instruction, and find a high filtering quality of $134/150 = 89.33\%$.

Miscellaneous For CoNLL2003, as there is already a miscellaneous type, we manually write an instruction to define the scope of miscellaneous. For MIT-Restaurant dataset, we combine "amenity", "hours", and "price" entity types. For MIT-Movie dataset, we combine "actor", "soundtrack", and "quote" entity types. Then we simply collect those types of entities to build the miscellaneous type for the benchmark. In the instruction, we include negations of miscellaneous as distractors to increase the difficulty in instruction-following.

The specific instructions used for instruction-following IE are listed in Table 7.

C Adaptive Supervision Scaling

In the application for IE, it's common to scale up the adaptive supervision (few-shot instances) to strengthen the model's IE ability. We plot such an example for CoNLL2003 in Figure 7 for transferring learning with different scales of supervision, from 5-shot to 320-shot. For comparison, we include the strongest NER baseline, NuNER, from the main experiment.

Task	Dataset	Instruction
Disamb.	CoNLL2003	The organization entity must be a subject of any active action in the context.
	BioBLP2004	The provided context must contain some descriptive information about the protein.
	Restaurant	The rating should describe a food or drink mentioned in the sentence.
Prefer.	SQuAD	Give the longest answer Give the shortest answer Give a concise answer
Misc.	CoNLL2003	Miscellaneous includes events, nationalities and products but not person, location or organization.
	Restaurant	Miscellaneous includes amenity, hours and price but not rating, dish, or location.
	Movie	Miscellaneous includes actor, soundtrack and quote but not director, opinion, or plot.

Table 7: The specific instructions used for instruction-following IE.

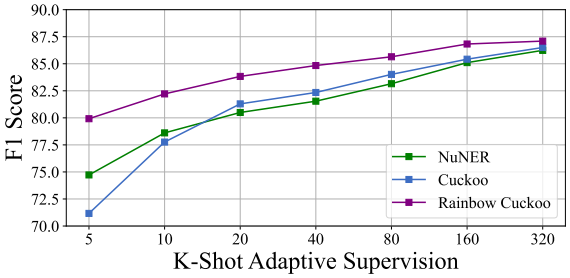


Figure 7: The scaling-up performance on adaptive supervision from CoNLL2003 of pre-trained IE models.

Rephrase	New Template/Label
Template	**User:** [Context] Instruction: Extract [Label] from the text above. **Assistant:** [Label]:
	User: List all [Label] entities: [Context] **Assistant:** Here are [Label] entities: 1.
Label	(CoNLL2003) Person → Name
	(BioBLP2004) DNA → Deoxyribonucleic acid
	(Restaurant) Rating → Recommendation
	(Movie) Genre → Category

Table 8: The template/label variants used for robustness testing.

The results demonstrate that Cuckoo can scale up similarly as NuNER, the in-domain transfer of NuNER shows its advantage under very weak supervision but is surpassed by Cuckoo when the adaptive supervision is enough for domain understanding. Finally, Rainbow Cuckoo consistently show advantages under different adaptive supervision scales.

D Robustness to Verbalization

As Cuckoo relies on prompts to perform different tasks. Its robustness to different verbalization of tasks and labels needs more emphasis. We select NER as an example and rephrase templates and labels in our experiments, which are listed in Table 8. We rerun the experiments with these modifications and find the NER performance is not significantly (defined as $p < 0.05$ in significance testing) different from the initial results. This indicates Cuckoo to be robustness to different verbalization styles.